

Evaluating Integrals and Solving Integral Equations with Laplace Transforms

A Self-Study Reference & Master Guide — Sections 2.7–2.8

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Part I — Theory & Worked Examples

Evaluating Integrals & Integral Equations at a Glance: Core Sub-Topics

- **The Laplace Method for Integrals (2.7):** a definite improper integral $\int_0^\infty f(t) dt$ is nothing more than the Laplace transform $F(s)$ evaluated at $s = 0$, because setting $s = 0$ erases the kernel e^{-st} and leaves the bare integral behind. Paired with the *division-by-t* property, this turns hard Calculus II integrals — the kind that have no elementary antiderivative — into one-line transform lookups.
- **The Laplace Method for Integral Equations (2.8):** an equation that hides its unknown $x(t)$ *inside* an integral of the form $\int_0^t K(t - \tau) x(\tau) d\tau$ is a *convolution* in disguise. The Convolution Theorem converts that integral into the ordinary product $K(s)X(s)$, collapsing the whole equation into linear algebra for $X(s)$.
- **One toolkit, two directions:** both methods rest on the same machinery — transform, exploit a structural property ($F(0)$, integration-in- s , or the convolution product), then either read off a number or invert back to $x(t)$. The Laplace transform is a bridge that carries calculus problems into algebra and walks the answers back.

Why a Transform Can Evaluate an Integral You Cannot Antidifferentiate

The deepest idea in this section is almost a sleight of hand. The definition of the transform,

$$F(s) = \int_0^\infty e^{-st} f(t) dt,$$

already *contains* an integral from 0 to ∞ . The factor e^{-st} is the only thing standing between $F(s)$ and the raw integral $\int_0^\infty f(t) dt$. Set $s = 0$ and that factor becomes $e^0 = 1$, so the kernel vanishes and $F(0)$ is the integral we wanted. This is why a problem that looks like Calculus II — an improper integral with no elementary antiderivative, such as $\int_0^\infty \frac{\sin t}{t} dt$ — can be cracked with a transform table instead of an antiderivative. We never antidifferentiate $\frac{\sin t}{t}$ (it has no elementary antiderivative); we transform it and evaluate at a point.

2.7 Evaluating Improper Integrals (Calculus II Style)

This first method answers a question that haunts every Calculus II student: *what do you do with an improper integral that has no elementary antiderivative?* The Laplace transform offers an escape hatch. Two properties do all the work — the evaluation-at- $s = 0$ identity, and the division-by- t (integration-in- s) property — and together they evaluate integrals that ordinary calculus cannot touch.

Property A — Evaluation at $s = 0$

If $\mathcal{L}\{f(t)\} = F(s)$ and the improper integral converges, then

$$\int_0^\infty f(t) dt = F(0).$$

The proof is immediate from the definition: $F(s) = \int_0^\infty e^{-st} f(t) dt$, so substituting $s = 0$ replaces e^{-st} by $e^0 = 1$, leaving $F(0) = \int_0^\infty f(t) dt$ exactly. The transform variable s is a “dial,” and turning it to zero recovers the undamped integral.

Property B — Division by t (Integration in the s -Domain)

If $\mathcal{L}\{f(t)\} = F(s)$, then dividing the time function by t corresponds to *integrating* the transform from s out to infinity:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\sigma) d\sigma.$$

This is the exact mirror of the familiar rule $\mathcal{L}\{t f(t)\} = -F'(s)$: multiplying by t *differentiates* the transform (with a sign), so dividing by t *integrates* it. The dummy variable σ sweeps from the current s out to ∞ ; the surviving variable is s .

Why do we need Property B at all? Because the integrals worth evaluating this way — like $\int_0^\infty \frac{\sin t}{t} dt$ — carry a stubborn $\frac{1}{t}$ that no table entry handles directly. Property B is the tool that absorbs that $\frac{1}{t}$: it tells us the transform of $\frac{f(t)}{t}$ without ever performing the division in the time domain. Once we have $\mathcal{L}\{f(t)/t\}$ as a clean function of s , Property A finishes the job by evaluating it at the right point.

The Recipe — Evaluating $\int_0^\infty \frac{f(t)}{t} e^{-at} dt$

Step 1 — Strip the integral down to a transform. Recognize that $\int_0^\infty e^{-at} \frac{f(t)}{t} dt$ is simply $\mathcal{L}\left\{\frac{f(t)}{t}\right\}$ evaluated at $s = a$. (When there is no exponential, $a = 0$ and you evaluate the limit as $s \rightarrow 0^+$.)

Step 2 — Find the base transform. Look up $F(s) = \mathcal{L}\{f(t)\}$ from the table.

Step 3 — Apply division-by- t . Compute $G(s) = \mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\sigma) d\sigma$. This integral in σ is the heart of the method.

Step 4 — Evaluate at the right point. Plug in $s = a$ to read off the numerical value: the answer is $G(a)$. If $a = 0$, take the limit $\lim_{s \rightarrow 0^+} G(s)$.

Worked Example 1: $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$

Evaluate the improper integral $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$, which has no elementary antiderivative.

Step 1 — Read the integral as a transform evaluated at a point. The factor e^{-t} is precisely the Laplace kernel e^{-st} with $s = 1$. So if we let $f(t) = \sin t$, the integral is the transform of $\frac{\sin t}{t}$ evaluated at $s = 1$:

$$\int_0^\infty \frac{e^{-t} \sin t}{t} dt = \int_0^\infty e^{-st} \frac{\sin t}{t} dt \Big|_{s=1} = \mathcal{L}\left\{\frac{\sin t}{t}\right\} \Big|_{s=1}.$$

Step 2 — Base transform. From the table, $\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$, so $F(\sigma) = \frac{1}{\sigma^2 + 1}$.

Step 3 — Apply division-by- t . By Property B,

$$G(s) = \mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{1}{\sigma^2 + 1} d\sigma.$$

The antiderivative of $\frac{1}{\sigma^2 + 1}$ is $\arctan \sigma$, so we evaluate the improper integral as a limit:

$$G(s) = \left[\arctan \sigma \right]_{\sigma=s}^{\sigma \rightarrow \infty} = \lim_{\sigma \rightarrow \infty} \arctan \sigma - \arctan s = \frac{\pi}{2} - \arctan s.$$

The combination $\frac{\pi}{2} - \arctan s$ is exactly $\operatorname{arccot} s = \arctan \frac{1}{s}$, so equivalently $G(s) = \arctan \frac{1}{s}$.

Step 4 — Evaluate at $s = 1$. Set $s = 1$ in $G(s)$:

$$\int_0^\infty \frac{e^{-t} \sin t}{t} dt = G(1) = \frac{\pi}{2} - \arctan 1 = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}.$$

The whole integral collapses to $\boxed{\frac{\pi}{4}}$ — found without ever antidifferentiating $\frac{\sin t}{t}$.

The Limit as $s \rightarrow 0$: A Free Bonus

Notice what $G(s) = \frac{\pi}{2} - \arctan s$ gives if we instead let $s \rightarrow 0^+$ — that is, if we drop the damping factor e^{-t} entirely and ask for the *undamped* integral:

$$\int_0^\infty \frac{\sin t}{t} dt = \lim_{s \rightarrow 0^+} \left(\frac{\pi}{2} - \arctan s \right) = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$$

This is the celebrated **Dirichlet integral**. The same $G(s)$ that gave us $\pi/4$ at $s = 1$ delivers the famous $\pi/2$ in the limit $s \rightarrow 0$ — a single transform answers a whole family of integrals, one for each value of the dial s . You will reprove this directly in Problem 1.

2.8 Solving Volterra Integral Equations

We now turn from *evaluating* integrals to *solving for an unknown trapped inside one*. A **Volterra integral equation of convolution type** has the form

$$x(t) = f(t) + \int_0^t K(t - \tau) x(\tau) d\tau,$$

where $f(t)$ (the forcing) and $K(t)$ (the *kernel*) are known, and $x(t)$ is the unknown we must find. The decisive observation is that the integral is a **convolution**: $\int_0^t K(t - \tau) x(\tau) d\tau = (K * x)(t)$. That single recognition unlocks everything.

The Convolution Theorem — The Master Key

The transform of a convolution is the *product* of the transforms:

$$\mathcal{L}\{(K * x)(t)\} = \mathcal{L}\left\{\int_0^t K(t - \tau) x(\tau) d\tau\right\} = K(s) X(s).$$

An operation that is genuinely hard in the time domain (the convolution integral) becomes ordinary multiplication in the s -domain. This is why integral equations succumb to Laplace methods: the integral that defines the equation simply turns into $K(s) X(s)$.

The Recipe — Solving the Convolution Equation

Step 1 — Transform every term. Apply $\mathcal{L}$ to $x(t) = f(t) + (K * x)(t)$. By the Convolution Theorem the integral becomes $K(s)X(s)$:

$$X(s) = F(s) + K(s) X(s).$$

Step 2 — Gather the $X(s)$ terms. Move $K(s)X(s)$ to the left:

$$X(s) - K(s) X(s) = F(s).$$

Step 3 — Factor out $X(s)$.

$$X(s) (1 - K(s)) = F(s).$$

Step 4 — Isolate $X(s)$ by dividing by the bracket:

$$X(s) = \frac{F(s)}{1 - K(s)}.$$

Step 5 — Decompose and invert via partial fractions to recover $x(t) = \mathcal{L}^{-1}\{X(s)\}$.

Worked Example 2: $x(t) = t^2 + \int_0^t \sin(t - \tau) x(\tau) d\tau$

Solve the Volterra equation $x(t) = t^2 + \int_0^t \sin(t - \tau) x(\tau) d\tau$.

Step 1 — Identify the pieces. The forcing is $f(t) = t^2$ and the kernel is $K(t) = \sin t$. From the table,

$$F(s) = \mathcal{L}\{t^2\} = \frac{2!}{s^3} = \frac{2}{s^3}, \quad K(s) = \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}.$$

Step 2 — Transform the equation. The integral is the convolution $(\sin * x)(t)$, which transforms to $K(s)X(s)$:

$$X(s) = \frac{2}{s^3} + \frac{1}{s^2 + 1} X(s).$$

Step 3 — Gather $X(s)$ on the left and factor.

$$X(s) - \frac{1}{s^2 + 1} X(s) = \frac{2}{s^3} \implies X(s) \left(1 - \frac{1}{s^2 + 1}\right) = \frac{2}{s^3}.$$

Step 4 — Combine the bracket over a common denominator. Write $1 = \frac{s^2 + 1}{s^2 + 1}$ so the two fractions share the denominator $s^2 + 1$, then subtract:

$$1 - \frac{1}{s^2 + 1} = \frac{s^2 + 1}{s^2 + 1} - \frac{1}{s^2 + 1} = \frac{(s^2 + 1) - 1}{s^2 + 1} = \frac{s^2}{s^2 + 1}.$$

So the equation becomes

$$X(s) \left(\frac{s^2}{s^2 + 1}\right) = \frac{2}{s^3}.$$

Step 5 — Multiply by the reciprocal to isolate $X(s)$. The reciprocal of $\frac{s^2}{s^2 + 1}$ is $\frac{s^2 + 1}{s^2}$:

$$X(s) = \frac{2}{s^3} \cdot \frac{s^2 + 1}{s^2} = \frac{2(s^2 + 1)}{s^5}.$$

The s -domain algebra is finished. The decomposition and inversion follow in the box below.

Worked Example 2 (continued): Decomposition and Inversion

We carry $X(s) = \frac{2(s^2 + 1)}{s^5}$ forward.

Step 6 — Set up the partial fraction decomposition. The denominator is the single repeated factor s^5 , so a complete decomposition assigns one term to each power from s^1 up to s^5 :

$$\frac{2(s^2 + 1)}{s^5} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s^4} + \frac{E}{s^5}.$$

Multiply both sides by s^5 to clear every denominator:

$$2(s^2 + 1) = A s^4 + B s^3 + C s^2 + D s + E.$$

Step 7 — Match coefficients power by power. Expand the left side as $2s^2 + 2$, i.e. $0 \cdot s^4 + 0 \cdot s^3 + 2 \cdot s^2 + 0 \cdot s + 2$, and equate each coefficient:

$$\underbrace{A=0}_{s^4}, \quad \underbrace{B=0}_{s^3}, \quad \underbrace{C=2}_{s^2}, \quad \underbrace{D=0}_{s^1}, \quad \underbrace{E=2}_{s^0}.$$

The formal decomposition therefore collapses to just two surviving terms:

$$X(s) = \frac{2}{s^3} + \frac{2}{s^5}.$$

(This matches what the eye sees if it splits the numerator directly: $\frac{2(s^2 + 1)}{s^5} = \frac{2s^2}{s^5} + \frac{2}{s^5} = \frac{2}{s^3} + \frac{2}{s^5}$. The full setup confirms there are no hidden $1/s$, $1/s^2$, or $1/s^4$ pieces.)

Step 8 — Invert term by term. Use $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$. For the first term, $\frac{2}{s^3} = 2 \cdot \frac{1}{s^3} \rightarrow 2 \cdot \frac{t^2}{2!} = 2 \cdot \frac{t^2}{2} = t^2$. For the second, $\frac{2}{s^5} = 2 \cdot \frac{1}{s^5} \rightarrow 2 \cdot \frac{t^4}{4!} = 2 \cdot \frac{t^4}{24} = \frac{t^4}{12}$:

$$x(t) = t^2 + \frac{t^4}{12}.$$

Step 9 — Verify. The forcing t^2 reappears as the leading term, so the convolution must supply exactly the $\frac{t^4}{12}$ correction. Indeed $\mathcal{L}\left\{\frac{t^4}{12}\right\} = \frac{1}{12} \cdot \frac{4!}{s^5} = \frac{2}{s^5}$, while $K(s)X(s) = \frac{1}{s^2 + 1} \cdot \frac{2(s^2 + 1)}{s^5} = \frac{2}{s^5}$ — the two agree, so $x(t) = t^2 + \frac{t^4}{12}$ solves the equation. ✓

Mind the Argument: $K(t - \tau)$, Not $K(\tau)$

The Convolution Theorem only applies when the kernel's argument is the *difference* $t - \tau$. An equation with $\int_0^t K(t - \tau) x(\tau) d\tau$ is a true convolution and transforms to $K(s)X(s)$. But $\int_0^t K(t) x(\tau) d\tau$ (kernel depending on t alone, pulled outside) or a kernel written as $K(t + \tau)$ is *not* a convolution — the product rule $K(s)X(s)$ does **not** apply. Always confirm the integrand has the clean $t - \tau$ structure before invoking the theorem.

Key Takeaway

Both halves of this section run on the transform's structure. To **evaluate** an improper integral, recognize it as $F(s)$ at a point — $s = 0$ for the raw integral, or $s = a$ when an e^{-at} rides along — and lean on **division-by- t** to absorb a $\frac{1}{t}$. To **solve** an integral equation, recognize the integral as a **convolution**, replace it with the product $K(s)X(s)$, and solve the one-line algebra

$$X(s) = \frac{F(s)}{1 - K(s)}.$$

Part II — Practice Set

Work each problem before turning to the complete solutions in Part III. Problems 1–4 evaluate improper integrals, Problems 5–8 solve Volterra integral equations (Problem 8 is integro-differential), and Problems 9–10 ask you to prove the two properties that power this section.

Problem 1: The Dirichlet Integral.

Evaluate $\int_0^\infty \frac{\sin t}{t} dt$ by finding $\mathcal{L}\{\sin t/t\}$ and taking the limit as $s \rightarrow 0^+$.

Problem 2: A Frullani-Type Integral.

Evaluate $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt$.

Problem 3: Evaluation at a Point.

Evaluate $\int_0^\infty t e^{-2t} \cos t dt$ by recognizing it as a transform evaluated at $s = 2$.

Problem 4: Division-by- t with Damping.

Evaluate $\int_0^\infty \frac{e^{-2t} \sin(3t)}{t} dt$.

Problem 5: A Volterra Equation.

Solve for $x(t)$:

$$x(t) = 1 + \int_0^t (t - \tau) x(\tau) d\tau.$$

Problem 6: A Volterra Equation.

Solve for $x(t)$:

$$x(t) = e^t + \int_0^t e^{t-\tau} x(\tau) d\tau.$$

Problem 7: A Volterra Equation.

Solve for $x(t)$:

$$x(t) = t + \int_0^t \sin(t - \tau) x(\tau) d\tau.$$

Problem 8: An Integro-Differential Equation.

Solve for $x(t)$, given $x(0) = 0$:

$$x'(t) = 1 - \int_0^t x(\tau) d\tau.$$

Problem 9 (Proof): The Integral Property of Laplace.

Prove the division-by- t property

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\sigma) d\sigma,$$

and deduce the evaluation corollary $\int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(\sigma) d\sigma$.

Problem 10 (Proof): Inversion of the Convolution Product.

Prove the Convolution Theorem — that the product $F(s) G(s)$ inverts to the convolution:

$$\mathcal{L}\{(f * g)(t)\} = F(s) G(s), \quad (f * g)(t) = \int_0^t f(\tau) g(t - \tau) d\tau.$$

Part III — Complete Solutions

Solution 1: The Dirichlet Integral $\int_0^\infty \frac{\sin t}{t} dt$.

Step 1 — Transform $\sin t/t$ via division-by- t . With $f(t) = \sin t$ and $F(\sigma) = \frac{1}{\sigma^2 + 1}$, Property B gives

$$\mathcal{L}\left\{\frac{\sin t}{t}\right\} = \int_s^\infty \frac{1}{\sigma^2 + 1} d\sigma = \left[\arctan \sigma\right]_s^\infty = \frac{\pi}{2} - \arctan s.$$

Step 2 — Recognize the raw integral as the $s \rightarrow 0^+$ limit. The undamped integral is $\mathcal{L}\{\sin t/t\}$ at $s = 0$:

$$\int_0^\infty \frac{\sin t}{t} dt = \lim_{s \rightarrow 0^+} \left(\frac{\pi}{2} - \arctan s\right) = \frac{\pi}{2} - 0 = \frac{\pi}{2}.$$

The Dirichlet integral equals $\frac{\pi}{2}$, obtained with no antiderivative of $\sin t/t$. ✓

Solution 2: $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt$.

Step 1 — Base transform. With $f(t) = e^{-t} - e^{-3t}$,

$$F(\sigma) = \frac{1}{\sigma + 1} - \frac{1}{\sigma + 3}.$$

Step 2 — Apply division-by- t . Integrate F from s to ∞ . The antiderivative is $\ln(\sigma + 1) - \ln(\sigma + 3) = \ln \frac{\sigma + 1}{\sigma + 3}$:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \left(\frac{1}{\sigma + 1} - \frac{1}{\sigma + 3}\right) d\sigma = \left[\ln \frac{\sigma + 1}{\sigma + 3}\right]_s^\infty.$$

As $\sigma \rightarrow \infty$, $\frac{\sigma + 1}{\sigma + 3} \rightarrow 1$ and $\ln 1 = 0$, so

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = 0 - \ln \frac{s + 1}{s + 3} = \ln \frac{s + 3}{s + 1}.$$

Step 3 — Evaluate at $s = 0$. The integral has no exponential damping ($a = 0$), so set $s = 0$:

$$\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt = \ln \frac{0 + 3}{0 + 1} = \ln 3.$$

(This is the Frullani value $\ln(b/a)$ with $a = 1$, $b = 3$.) ✓

Solution 3: $\int_0^\infty t e^{-2t} \cos t dt$.

Step 1 — Recognize the transform. The integral is $\mathcal{L}\{t \cos t\}$ evaluated at $s = 2$, since the e^{-2t} is the kernel with $s = 2$:

$$\int_0^\infty t e^{-2t} \cos t dt = \mathcal{L}\{t \cos t\}|_{s=2}.$$

Step 2 — Find $\mathcal{L}\{t \cos t\}$ by the multiply-by- t rule. Using $\mathcal{L}\{t f(t)\} = -F'(s)$ with $\mathcal{L}\{\cos t\} = \frac{s}{s^2 + 1}$:

$$\mathcal{L}\{t \cos t\} = -\frac{d}{ds} \left(\frac{s}{s^2 + 1} \right) = -\frac{(1)(s^2 + 1) - s(2s)}{(s^2 + 1)^2} = -\frac{1 - s^2}{(s^2 + 1)^2} = \frac{s^2 - 1}{(s^2 + 1)^2}.$$

Step 3 — Evaluate at $s = 2$.

$$\int_0^\infty t e^{-2t} \cos t \, dt = \frac{2^2 - 1}{(2^2 + 1)^2} = \frac{3}{25}.$$

✓

Solution 4: $\int_0^\infty \frac{e^{-2t} \sin(3t)}{t} \, dt.$

Step 1 — Read as a transform at $s = 2$. With $f(t) = \sin 3t$, the integral is $\mathcal{L}\{\sin 3t/t\}$ at $s = 2$.

Step 2 — Division-by- t . Here $F(\sigma) = \frac{3}{\sigma^2 + 9}$, whose antiderivative is $\arctan \frac{\sigma}{3}$:

$$\mathcal{L}\left\{\frac{\sin 3t}{t}\right\} = \int_s^\infty \frac{3}{\sigma^2 + 9} \, d\sigma = \left[\arctan \frac{\sigma}{3} \right]_s^\infty = \frac{\pi}{2} - \arctan \frac{s}{3}.$$

Step 3 — Evaluate at $s = 2$. Using $\frac{\pi}{2} - \arctan x = \arctan \frac{1}{x}$ with $x = \frac{2}{3}$:

$$\int_0^\infty \frac{e^{-2t} \sin 3t}{t} \, dt = \frac{\pi}{2} - \arctan \frac{2}{3} = \arctan \frac{3}{2} \approx 0.9828.$$

✓

Solution 5: $x(t) = 1 + \int_0^t (t - \tau) x(\tau) \, d\tau.$

Step 1 — Identify and transform. The kernel is $K(t) = t$ with $K(s) = \frac{1}{s^2}$, and $f(t) = 1$ with $F(s) = \frac{1}{s}$. The integral is the convolution $(t * x)(t) \rightarrow K(s)X(s)$:

$$X(s) = \frac{1}{s} + \frac{1}{s^2} X(s).$$

Step 2 — Gather and factor.

$$X(s) - \frac{1}{s^2} X(s) = \frac{1}{s} \implies X(s) \left(1 - \frac{1}{s^2} \right) = \frac{1}{s}.$$

Step 3 — Common denominator in the bracket.

$$1 - \frac{1}{s^2} = \frac{s^2 - 1}{s^2}, \quad \text{so} \quad X(s) \left(\frac{s^2 - 1}{s^2} \right) = \frac{1}{s}.$$

Step 4 — Multiply by the reciprocal and cancel.

$$X(s) = \frac{1}{s} \cdot \frac{s^2}{s^2 - 1} = \frac{s}{s^2 - 1}.$$

Step 5 — Invert. Recognize $\frac{s}{s^2 - 1}$ as the transform of $\cosh t$:

$$x(t) = \cosh t.$$

Check. $1 + \int_0^t (t - \tau) \cosh \tau \, d\tau = 1 + (\cosh t - 1) = \cosh t = x(t)$. ✓

Solution 6: $x(t) = e^t + \int_0^t e^{t-\tau} x(\tau) \, d\tau$.

Step 1 — Identify and transform. Here $K(t) = e^t$ with $K(s) = \frac{1}{s-1}$, and $f(t) = e^t$ with $F(s) = \frac{1}{s-1}$. The integral is $(e^t * x)(t) \rightarrow K(s)X(s)$:

$$X(s) = \frac{1}{s-1} + \frac{1}{s-1} X(s).$$

Step 2 — Gather and factor.

$$X(s) \left(1 - \frac{1}{s-1} \right) = \frac{1}{s-1}.$$

Step 3 — Common denominator in the bracket. Write $1 = \frac{s-1}{s-1}$:

$$1 - \frac{1}{s-1} = \frac{(s-1)-1}{s-1} = \frac{s-2}{s-1}, \quad \text{so} \quad X(s) \left(\frac{s-2}{s-1} \right) = \frac{1}{s-1}.$$

Step 4 — Multiply by the reciprocal and cancel. The factors of $(s-1)$ cancel:

$$X(s) = \frac{1}{s-1} \cdot \frac{s-1}{s-2} = \frac{1}{s-2}.$$

Step 5 — Invert.

$$x(t) = e^{2t}.$$

Check. $e^t + \int_0^t e^{t-\tau} e^{2\tau} \, d\tau = e^t + e^t \int_0^t e^{\tau} \, d\tau = e^t + e^t(e^t - 1) = e^{2t} = x(t)$. ✓

Solution 7: $x(t) = t + \int_0^t \sin(t - \tau) x(\tau) \, d\tau$.

Step 1 — Identify and transform. Kernel $K(t) = \sin t$ with $K(s) = \frac{1}{s^2 + 1}$; forcing $f(t) = t$ with $F(s) = \frac{1}{s^2}$:

$$X(s) = \frac{1}{s^2} + \frac{1}{s^2 + 1} X(s).$$

Step 2 — Gather, factor, and combine the bracket.

$$X(s) \left(1 - \frac{1}{s^2 + 1} \right) = \frac{1}{s^2}, \quad 1 - \frac{1}{s^2 + 1} = \frac{s^2}{s^2 + 1}, \quad \text{so} \quad X(s) \frac{s^2}{s^2 + 1} = \frac{1}{s^2}.$$

Step 3 — Multiply by the reciprocal.

$$X(s) = \frac{1}{s^2} \cdot \frac{s^2 + 1}{s^2} = \frac{s^2 + 1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}.$$

Step 4 — Invert term by term. Using $\mathcal{L}^{-1}\{1/s^2\} = t$ and $\mathcal{L}^{-1}\{1/s^4\} = \frac{t^3}{3!} = \frac{t^3}{6}$:

$$x(t) = t + \frac{t^3}{6}.$$

Check. The forcing t reappears, and the convolution must supply $\frac{t^3}{6}$; indeed $K(s)X(s) = \frac{1}{s^2 + 1} \cdot \frac{s^2 + 1}{s^4} = \frac{1}{s^4} \rightarrow \frac{t^3}{6}$. ✓

Solution 8: $x'(t) = 1 - \int_0^t x(\tau) d\tau$, $x(0) = 0$.

Step 1 — Transform each term. The derivative rule gives $\mathcal{L}\{x'\} = sX(s) - x(0) = sX(s)$ since $x(0) = 0$. The accumulation integral $\int_0^t x(\tau) d\tau$ is the convolution of x with the constant 1, so it transforms to $\frac{X(s)}{s}$; and $\mathcal{L}\{1\} = \frac{1}{s}$:

$$sX(s) = \frac{1}{s} - \frac{X(s)}{s}.$$

Step 2 — Gather the $X(s)$ terms. Add $\frac{X(s)}{s}$ to both sides:

$$sX(s) + \frac{X(s)}{s} = \frac{1}{s}.$$

Step 3 — Factor out $X(s)$.

$$X(s) \left(s + \frac{1}{s} \right) = \frac{1}{s}.$$

Step 4 — Common denominator in the bracket.

$$s + \frac{1}{s} = \frac{s^2 + 1}{s}, \quad \text{so} \quad X(s) \left(\frac{s^2 + 1}{s} \right) = \frac{1}{s}.$$

Step 5 — Multiply by the reciprocal and cancel. The factors of s cancel:

$$X(s) = \frac{1}{s} \cdot \frac{s}{s^2 + 1} = \frac{1}{s^2 + 1}.$$

Step 6 — Invert.

$$x(t) = \sin t.$$

Check. $x(0) = \sin 0 = 0$ ✓; and $x'(t) = \cos t$, while $1 - \int_0^t \sin \tau d\tau = 1 - (1 - \cos t) = \cos t$. ✓

Solution 9 (Proof): The Integral Property of Laplace.

Goal. Show $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\sigma) d\sigma$.

Step 1 — Start from the right-hand side. Replace $F(\sigma)$ by its defining integral, $F(\sigma) = \int_0^\infty e^{-\sigma t} f(t) dt$, and write the result as a double integral:

$$\int_s^\infty F(\sigma) d\sigma = \int_s^\infty \left(\int_0^\infty e^{-\sigma t} f(t) dt \right) d\sigma.$$

Step 2 — Exchange the order of integration. Both variables are independent over the region $\sigma \in [s, \infty)$, $t \in [0, \infty)$, so (by Fubini, valid where the transform converges absolutely) we integrate in σ first:

$$\int_s^\infty F(\sigma) d\sigma = \int_0^\infty f(t) \left(\int_s^\infty e^{-\sigma t} d\sigma \right) dt.$$

Step 3 — Evaluate the inner σ -integral. Treating $t > 0$ as constant,

$$\int_s^\infty e^{-\sigma t} d\sigma = \left[-\frac{1}{t} e^{-\sigma t} \right]_{\sigma=s}^{\sigma \rightarrow \infty} = 0 - \left(-\frac{1}{t} e^{-st} \right) = \frac{e^{-st}}{t}.$$

Substituting this back into the t -integral,

$$\int_s^\infty F(\sigma) d\sigma = \int_0^\infty f(t) \frac{e^{-st}}{t} dt = \int_0^\infty e^{-st} \frac{f(t)}{t} dt.$$

Step 4 — Recognize the transform. The last integral is, by definition, $\mathcal{L}\left\{\frac{f(t)}{t}\right\}$:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\sigma) d\sigma. \quad \blacksquare$$

Corollary (evaluation at $s = 0$). Setting $s = 0$ makes $e^{-st} = 1$ on the left, so $\mathcal{L}\{f/t\}|_{s=0} = \int_0^\infty \frac{f(t)}{t} dt$, giving

$$\int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(\sigma) d\sigma.$$

This is exactly the engine behind Solutions 1, 2, and 4.

Solution 10 (Proof): Inversion of the Convolution Product.

Goal. Show $\mathcal{L}\{(f * g)(t)\} = F(s)G(s)$.

Step 1 — Write the product as a double integral. Use a distinct dummy variable in each transform so they do not collide:

$$F(s)G(s) = \left(\int_0^\infty e^{-s\tau} f(\tau) d\tau \right) \left(\int_0^\infty e^{-s\rho} g(\rho) d\rho \right) = \int_0^\infty \int_0^\infty e^{-s(\tau+\rho)} f(\tau) g(\rho) d\rho d\tau.$$

Step 2 — Substitute $t = \tau + \rho$ in the inner integral. Hold τ fixed and let $t = \tau + \rho$, so $\rho = t - \tau$ and $d\rho = dt$. As ρ runs from 0 to ∞ , the new variable t runs from τ to ∞ :

$$F(s)G(s) = \int_0^\infty \int_\tau^\infty e^{-st} f(\tau) g(t - \tau) dt d\tau.$$

Step 3 — Swap the order of integration. The region is $0 \leq \tau \leq t < \infty$. Reading it with t outermost, τ ranges from 0 to t :

$$F(s)G(s) = \int_0^\infty e^{-st} \left(\int_0^t f(\tau) g(t - \tau) d\tau \right) dt.$$

Step 4 — Recognize the convolution. The inner integral is precisely $(f * g)(t)$, so the outer integral is its Laplace transform:

$$F(s)G(s) = \int_0^\infty e^{-st} (f * g)(t) dt = \mathcal{L}\{(f * g)(t)\}. \quad \blacksquare$$

Read backwards, this says the *product* $F(s)G(s)$ inverts to the *convolution* $f * g$ — the identity that turned every Volterra equation in this section into one line of algebra.

Closing Takeaway

The two properties proved above are the spine of the entire section. The **integral property** ($\mathcal{L}\{f/t\} = \int_s^\infty F d\sigma$, evaluated at a point) lets a transform table *evaluate* improper integrals that have no elementary antiderivative. The **convolution theorem** ($\mathcal{L}\{f * g\} = F G$) lets the same table *solve* for an unknown buried inside an integral. Whether the goal is a number or a function, the move is identical: cross the bridge into the s -domain, do easy algebra, and carry the answer back.